



Pharmacokinetic and metabolite profile of orally administered anemoside B4 in rats with an improved exposure in formulations of rectal suppository

Yaqing Ye^{a,b,c,1}, Mingzhen Xue^{b,d,1}, Xiaoting Tian^b, Hongwei Gao^e, Pei Hu^b, Linwei Wang^{b,f}, Jing Leng^{a,c,*}, Yaru Xue^{b,**}, Chenggang Huang^{b,***}

^a Guangxi Key Laboratory of Translational Medicine for Treating High-Incidence Infectious Diseases with Integrative Medicine, Guangxi University of Chinese Medicine, Nanning, 530000, China

^b Shanghai Institute of Materia Medica, Chinese Academy of Sciences, Shanghai, 201203, China

^c Department of Microbiology & Immunology, Guangxi Chinese Medicine University, Nanning, 530000, China

^d Jiangsu Key Laboratory for Functional Substances of Chinese Medicine, Nanjing University of Chinese Medicine, Nanjing, 210023, China

^e College of Pharmacy, Guangxi University of Chinese Medicine, Nanning, 530000, China

^f College of Chemistry and Chemical Engineering, Shanghai University of Engineering Science, Shanghai, 201620, China

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ABSTRACT

Ethnopharmacological relevance: *Pulsatilla chinensis* (Bunge) Regel is a traditional Chinese herbal medicine used to treat intestinal amebiasis, malaria, vaginal trichomoniasis, and bacterial infections. Anemoside B4 (AB4), a pentacyclic triterpenoid saponin, is one of the primary bioactive substances in *Pulsatilla chinensis* (Bunge) Regel, and gavage administration of AB4 to animals has been demonstrated to exhibit anticancer, anti-inflammatory, and antiviral actions. However, AB4 exposure in plasma is very low after oral administration, and the biotransformation of AB4 in vivo after oral administration remains unknown.

Aim of the study: The reason for conducting this research was to explore the metabolite profile of AB4 in rats following oral administration. Additionally, we aimed to develop an appropriate extravascular formulation to increase the exposure and duration of AB4 in vivo.

Materials and methods: A well-validated HPLC-QQQ-MS/MS method was used for the quantification of AB4 in plasma and was further applied to evaluate and compare the pharmacokinetic properties of AB4 dissolved in a saline solution and AB4 formulations in a rectal suppository or enteric capsule. Reliable UHPLC coupled to Q-Exactive Plus high-resolution MS was used to identify the metabolites in rat plasma, bile, urine, and faeces.

Results: AB4 was extensively metabolized, and a total of 29 metabolites were identified. The primary metabolic routes included deglycosylation, oxidation, dehydrogenation, reduction, sulfation, hydration, acetylation, and glucuronidation. The pharmacokinetic comparison showed that both the rectal suppository and enteric capsule increased the exposures of AB4 and one of its active metabolites, 23-hydroxybetulinic acid (23-HA). Notably, rectal suppositories increased systemic AB4 exposure ($AUC_{0-\infty}$) by approximately 49 and 28 times higher than that of the AB4 saline solution and enteric capsules, respectively. The $t_{1/2}$ of AB4 was extended to approximately 7 h after rectal administration compared to 2 h after oral administration.

Abbreviations: *P. chinensis*, *Pulsatilla chinensis* (Bunge) Regel; AB4, anemoside B4; AA3, anemoside A3; 23-HA, 23-hydroxybetulinic acid; PK, pharmacokinetic; SD, Sprague–Dawley; i.p., intraperitoneal; CKD, chronic kidney disease; IS, internal standard; HPLC-QQQ-MS/MS, High-performance liquid chromatography triple quadrupole tandem mass spectrometry; ESI, electron spray ionization; IACUC, Institutional Animal Care and Use Committee; iv, intravenous; AUC, area under the curve; C_{max} , maximum plasma concentrations; T_{max} , time of maximum concentration; $t_{1/2}$, terminal half-life.

* Corresponding author. Guangxi Key Laboratory of Translational Medicine for Treating High-Incidence Infectious Diseases with Integrative Medicine, Guangxi University of Chinese Medicine, Nanning, 530000, China.

** Corresponding author.

*** Corresponding author.

E-mail addresses: yyq961005@163.com (Y. Ye), 1016543984@qq.com (M. Xue), zhghydxxt@163.com (X. Tian), gaohongwei06@126.com (H. Gao), hupei@crjz.com (P. Hu), wlw11101316@163.com (L. Wang), l986771558@163.com (J. Leng), xueyaru3@simm.ac.cn (Y. Xue), cghsimm@126.com (C. Huang).

¹ These authors contributed equally to this work.

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Conclusion: Overall, our study demonstrated that the mismatched exposure-response relationship of AB4 could result from extensive metabolism in the gastrointestinal and circulatory systems. Thus, a rectal suppository could be an alternative formulation of AB4 to obtain both higher and longer exposure.

1. Introduction

Pulsatilla chinensis (Bunge) Regel (*P. chinensis*) is a well-known traditional Chinese medicine (TCM) and has been used to treat various infectious diseases, such as malaria, intestinal amebiasis, and malignant tumours (Li et al., 2020). Triterpene saponins (TSs) such as anemoside B4 (AB4), anemoside A3 (AA3), and 23-hydroxybetulinic acid (23-HA) are the primary chemical constituents of *P. chinensis* (Fig. 1). AB4 is the most abundant and is utilized as a quality control marker for *P. chinensis* (Jin et al., 2017; Ling et al., 2016; Ouyang et al., 2014). As a pure component, AB4 has also attracted research interest for its pharmacological activities, such as antiviral (Hu et al., 2016), anti-inflammatory, and immunoregulatory activities (Hu et al., 2009; Zhou et al., 2019).

Recently, oral administration of AB4 has been found to exert great therapeutic potential for some chronic diseases, such as chronic kidney disease (Gong et al., 2019) (Zhou et al., 2019) and hyperuricaemia (unpublished data). Nevertheless, our previous pharmacokinetic (PK) study of the extracted *P. chinensis* saponin mixture (AB4 content: 94.6%), which was orally administered to Sprague–Dawley (SD) rats, indicated that the exposure of AB4 was comparatively low in the portal vein, liver, and peripheral plasma compared to that of its probable metabolite, 23-HA (Guo et al., 2018). This study indicated that the oral dosage of AB4 could undergo extensive metabolism mediated by gut microbiota and hepatocytes. However, based on the available studies, the absolute bioavailability and metabolite profile of AB4 in vivo after oral administration have not been fully investigated. Considering the growing beneficial effects of AB4 for chronic diseases requiring long-term administration, it is necessary to deduce the PK behaviour and metabolic profile of AB4 in vivo after oral administration.

Two investigations were previously done to look into the in vivo and in vitro metabolites of AB4. After intraperitoneal (i.p.) injection of AB4 (50 mg/kg), the in vivo research revealed eight metabolites in rat plasma, urine, and faeces, with deglycosylation and hydroxylation as the predominant metabolic routes (He et al., 2017). The in vitro study with AB4 incubated in the rat's small and large intestine microflora found ten metabolites, including 23-HA and AA3. The main pathways involved oxygenation and/or deglycosylation (Wan et al., 2017). The above results provided some valuable information for AB4 biotransformation but could not indicate the actual metabolic profile of AB4 after oral administration. The metabolic profile is important for new drug development and remains to be investigated for AB4.

AB4 is a triterpene saponin containing a variety of glycosidic bonds, and the glycoside is readily hydrolysed in an acidic environment. This structural characteristic is one of the reasons for the low oral bioavailability of this type of compound. For instance, ginsenosides undergo metabolic pathways such as deglycosylation, hydration, and dehydration in gastric fluid, which are one of the reasons for their low

bioavailability (Huang et al., 2022). Therefore, due to this structural property, we developed a formulation of AB4 enteric capsules to avoid hydrolysis of AB4 by gastric acid. In addition, previous laboratory data suggested that an oral dosage of AB4 could undergo extensive metabolism mediated by gut microbiota and hepatocytes, and we developed a rectal suppository to partially avoid the first-pass effect of the liver and the biotransformation by small and large intestine microflora.

In the current investigation, we described the PK characteristics of AB4 following oral and intravenous administration and obtained absolute bioavailability information. Furthermore, comprehensive metabolite identification of AB4 after oral administration was conducted for rat urine, blood, faeces, and bile by UHPLC-Q Exactive Plus MS. Finally, for the clinical benefit of AB4 in patients with chronic diseases such as CKD or hyperuricaemia, we developed an AB4 formulation of the enteric capsule and rectal suppository, whose PK properties were investigated and compared in rats.

2. Materials and methods

2.1. Materials and reagents

Standards of AB4 (98% purity), AA3 (98% purity), and 23-HA (98% purity) were purchased from Shanghai Shidande Co. Ltd. (Shanghai, China). From Tokyo Chemical Industries Co. Ltd., digoxin (98% purity) was acquired as the internal standard (IS) (Tokyo, Japan). The oral delivery system, medication filling apparatus, and the enteric capsule of rats coated with polyacrylic resin material were bought from Yuyan Instruments (Shanghai, China). The rectal suppository was kindly provided by Professor Gao Hongwei, School of Pharmacy, Guangxi University of Chinese Medicine (Nanning, China). AB4 rectal suppositories are mainly composed of AB4 standard and fatty acid glycerides. The preparation process can be seen in the supplementary materials. Thermo Fisher Scientific Co. Ltd. (Waltham, Massachusetts, USA) supplied formic acid and acetonitrile (HPLC grade). The remaining analytical-grade chemicals were purchased from Sigma-Aldrich (St. Louis, MO, USA) or Sinopharm Chemical Reagent Co. Ltd. (Shanghai, China). A Milli-Q System (Millipore, Billerica, MA, USA) was used to prepare purified water for the mobile phase.

2.2. Instrumentations and analytical conditions

2.2.1. High performance liquid chromatography triple quadrupole tandem mass spectrometry (HPLC-QQQ-MS/MS) conditions for quantitative analysis

An HPLC-QQQ-MS/MS bioanalytical method was created and validated in our lab for the measurement of AB4 in rat plasma (Guo et al., 2018). An Agilent 6460 triple quadrupole mass spectrometer (Agilent

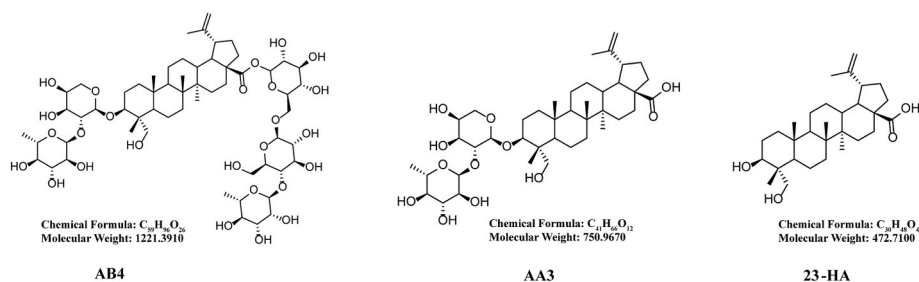


Fig. 1. Chemical structures of AB4, AA3, and 23-HA from *Pulsatilla chinensis* saponins.

Technologies, Palo Alto, CA, USA) outfitted with an Agilent 1260 Series liquid chromatography system was used to analyze the measurement of AB4 and 23-HA in biological samples. The quantification of the analytes and IS was optimized in the negative multiple reaction monitoring (MRM) mode utilizing the ion transitions for AB4, AA3, 23-HA, and digoxin (IS) of 1219.5 → 749.4, m/z 795.5 → 749.4, m/z 471.4 → 471.4 and m/z 825.3 → 779.1. Chromatographic separation was conducted using an ACE C18 column (100 mm × 2.1 mm, 3.0 μ m, Advanced Chromatography Technologies, USA) connected to an ACE 3 SuperC18 guard column. Solvent A (water with 0.1% formic acid) and Solvent B (acetonitrile with 0.1% formic acid) were used as the mobile phase for the PK profile determination. The instrument settings were a column temperature of 40 °C and a flow rate of 0.35 mL/min. The gradient solvent system was as follows: 3.5 min, 26% B; 4.0 min, 40% B; 5.0 min, 60% B; 6.5 min, 60% B; 6.6 min, 70% B; 7.5 min, 70% B; 7.6 min, 95% B; 8.0 min, 95% B; 8.1 min, 26% B; and 10.0 min, 26% B.

2.2.2. UHPLC-Q Exactive Plus MS conditions for metabolite identification

An ACE Excel 3 C18-AR column (3 μ m, 100 mm × 2.1 mm) was used for separation on a Vanquish UHPLC system (Thermo, Waltham, MA, USA) at 40 °C. The 265 nm wavelength was chosen. The mobile phases consisted of water containing 0.1% formic acid and 5 mM ammonium acetate (A) and acetonitrile containing 0.1% formic acid (B). The gradient elution process went as follows: 0 min, 10% B; 1 min, 10% B; 9 min, 40% B; 13 min, 85% B; 16 min, 95% B; 18 min, 95% B; 18.1 min, 10% B; and 20 min, 10% B. A flow rate of 0.5 mL/min was used.

Using heated electron spray ionization (ESI⁺), Q-Exactive MS (Thermo) produced the mass data. Mass data were generated by Q-Exactive MS (Thermo) using heated electron spray ionization (ESI⁺). The following mass parameters were chosen: ion spray voltage of 3.2 kV; ion-source temperature of 320 °C; sheath gas of 45 units; auxiliary gas of 10 units; and normalized collision energies of 20, 40, and 60 V. HRMS scanning was applied at the full MS scan in the mass range of m/z 150–2250 with a resolution of 35,000, and MS2 was applied at a resolution of 17,500. Xcalibur software was utilized to process all of the mass data.

2.3. Animals

The animal study followed the recommendations of the Shanghai Institute of Materia Medica, Chinese Academy of Sciences' Institutional Animal Care and Use Committee (IACUC) (Shanghai, China). It was performed in accordance with all applicable standard operating procedures. SD rats (male, 190–290 g) were purchased from the Shanghai SLAC Laboratory Animal Co. Ltd. [Shanghai, China, Certificate No. SCXK (Zhe) 2019-0001]. They were kept in an environment with regulated lighting (12 h of strong light, 12 h of darkness), relative humidity of 50% ± 10%, and temperature of 22 ± 2 °C. The rats had unrestricted access to water during their 12-h fast.

2.3.1. Pharmacokinetic (PK) study

Twenty-eight male SD rats were used in the PK study and were randomly divided into four groups (intravenous, saline solution, rectal suppository, and enteric capsule, $n = 7$). The rats received a 100 μ mol (122 mg/kg) dose of AB4 with the formulations for saline solution, rectal suppository, and enteric capsule administration. AB4 in saline was injected intravenously into the remaining seven rats at a dose of 1 μ mol (1.22 mg/kg). The rats were anaesthetized with isoflurane, and then blood (0.5 mL) was collected in 1.5 mL tubes containing heparin sodium by means of jugular vein sampling at each time point at 0.167, 0.5, 1, 2, 4, 8, 12, 24, 48, and 72 h (saline solution, rectal suppository, and enteric capsule) or 0.083, 0.167, 0.5, 1, 2, 4, 8, 12, and 24 h (intravenous) after dosing. The samples were immediately vortexed, centrifuged, and the plasma was extracted from the supernatant. Prior to analysis, all samples were kept at −80 °C.

2.3.2. Metabolite identification study

The 12 male SD rats were split into 3 groups (groups 1, 2, and 3, $n = 4$) at random. Prior to giving the dosage solution to the rats, blank plasma, urine, faeces, and bile were collected. Rats in group 1 received a single dosage of AB4 (100 mg/kg), and blood samples were collected by jugular vein collection in 1.5 mL tubes containing heparin sodium at 0.167, 0.5, 1, 1.5, 2, 4, 6, 8, 24, 48, and 72 h. The plasma was extracted from the blood samples after they had been centrifuged at 3500 rpm for 10 min. Moreover, faeces samples were obtained at 0–24, 24–48, and 48–72 h after treating rats in group 2 and urine samples were taken at 0–8, 8–24, 24–48, and 48–72 h. At the time of each urine collection, a 10 mL amount of methanol-water (1:1, v/v) was used to wash the cage. During the bile excretion experiment, four bile duct cannulated (BDC) SD rats were kept separately in metabolism cages. Before the dosage as well as at 0–4, 4–8, 8–24, 24–48, and 48–72 h after the dose, bile was collected (The method for bile collection was provided in the supplementary materials). Prior to analysis, all samples were weighed and kept at −20 °C.

2.4. Preparation of standard solution and quality control samples

Stock and working solutions of AB4, AA3, 23-HA, and digoxin (IS) were prepared in acetonitrile at 1 mg/mL. Calibration curve samples were prepared by spiking the working solutions in blank plasma (100 μ L) to yield final concentrations of 3, 10, 30, 100, 300, 1000, and 3000 ng/mL, along with 50 ng/mL for the IS. The same procedure was used to create three concentration levels of quality control (QC) samples that contained three analytes (8, 240, and 2400 ng/mL for AB4 and 23 HA and AA3 respectively) and the IS (5 μ g/mL). While not in use, all of the working and stock solutions were kept at −20 °C.

2.5. Sample preparation

2.5.1. Pharmacokinetic samples

The liquid-liquid extraction technique was used to plasma sample treatment. An aliquot of 100 μ L rat plasma was spiked with 10 μ L of IS solution (50 μ g/mL digoxin) and subsequently extracted with a volume of 1 mL of a mixture of ethyl acetate and isopropanol (v/v, 50:50). The upper organic layer was dried in a vacuum drying machine after vortex mixing for 10 min and centrifuging at 13000 rpm for 10 min. The residue was dissolved in 100 μ L of the mobile phase, which was made up of water, acetonitrile, and acetic acid in the proportion 74:26:0.1, and centrifuged at 13000 rpm for 10 min, and 10 μ L of the supernatant was subjected to HPLC-MS/MS for quantitative analysis.

2.5.2. Metabolite identification samples

The plasma samples taken at 0 (predose), 0.167, 0.5, 1, 1.5, 2, 4, 6, 8, and 24 h postdose were selected for the PK study. Plasma samples were pooled in accordance with the AUC-pooling principle (Hop et al., 1998). Urine and bile samples were pooled by combining equal percentages of the total volume excreted from each rat. Each faecal sample was added to three times the weight of acetonitrile/water (50:50, v/v) to obtain the homogenate. An equal percentage by weight of each homogenate (0–24, 24–48, and 48–72 h) was pooled from the same sex to obtain one pooled faecal sample. The pooled plasma, urine, bile, and faeces samples were processed in our lab using the same procedure as in a prior research, and bile samples were treated in the same manner as urine samples (Hu et al., 2021).

2.6. Data analysis

The Winnonlin software (Pharsight 6.2, NC, USA) was used to process the PK data, and the non-compartmental method was used to estimate the PK parameters. Metabolites in plasma, urine, bile, and faeces were discovered using Compound Discovery 3.3 software. The concentration-time curve was produced using GraphPad Prism 8.0

(GraphPad Software, USA). All of the results are expressed as the mean $\pm$ SD.

3. Results

3.1. Pharmacokinetic study of AB4 after oral and intravenous administration

Previously, the HPLC-QQQ-MS/MS bioanalytical method used for AB4 quantitation was established and validated in our laboratory (Guo et al., 2018). The PK study was conducted in rats with an oral dose (*po*) of 122 mg/kg AB4 solution and an intravenous dose (*iv*) of 1.22 mg/kg AB4 solution in saline. The mean area under the curve (AUC) of AB4 using the two administration routes is presented in Fig. 2. The calculated PK parameters are summarized in Table 1.

The maximum plasma concentrations (C_{max}) of AB4 were 94.47 ± 28.99 ng/mL (*po*) and 13615 ± 838.18 ng/mL (*iv*), with time of maximum concentration observed (T_{max}) values of 2.33 ± 1.53 h and 0.08 ± 0.00 h, respectively. Meanwhile, the terminal half-life ($t_{1/2}$) values were 2.93 ± 1.04 h (*po*) and 1.06 ± 0.070 h (*iv*). The mean area under the concentration–time curve ($AUC_{0-\infty}$) values were 592.55 ± 57.78 ng/mL·h (*po*) and 15671.81 ± 2001.37 ng/mL·h (*iv*), with AUC/dose values of 4.86 ng/mL·h (*po*) and 12845.75 ng/mL·h (*iv*); these results suggested that the exposure to AB4 after oral administration was quite low. The bioavailability was extremely low and less than 1%.

3.2. Metabolite identification of AB4

Using UHPLC-Q Exactive Plus MS, the metabolites of AB4 in rat plasma, urine, bile, and faeces were analysed. Metabolites were identified and provisionally defined following the study of the chromatographic patterns of the biological samples containing drugs and their matching mass spectral data. Table 2 summarizes the route, retention periods, and mass spectrum information.

3.2.1. The reference standards of AB4 and its metabolites

Parent AB4. Under the experimental conditions used in this study, the retention time of AB4 ($C_{59}H_{96}O_{26}$) was 8.76 min, and a molecular ion ($[M + HCOO]^-$) at m/z 1265.6172 was obtained in full-scan MS. In the product ion scan, the primary fragment ions obtained were m/z 749.4473, 469.1569, 323.0974, 247.0824, 161.0437, and 101.0231 (Fig. 3a). The fragment ion at m/z 749.4473 was formed by loss of sugar ring A. The fragment ion at 469.1569 was produced by the removal of two hydrogens from sugar ring A.

AA3. AA3 was the ester bond hydrolysate of AB4 with a retention time of 12.09 min ($[M + HCOO]^-$ m/z 795.4536). The major fragment ions of AA3 were m/z 749.4496, 603.3889, 471.3483, and 101.0241. The fragment ions at m/z 603.3889 were formed by the loss of sugar ring B1. The fragment ions at m/z 471.3483 were formed by the loss of sugar ring B and had the same exact mass as 23-HA. The fragment ions m/z

Table 1

Pharmacokinetic parameters of AB4 after oral solution and intravenous administration. (n = 7, mean $\pm$ SD).

Parameters	AB4-oral solution (122 mg/kg)	AB4-intravenous (1.22 mg/kg)
$T_{1/2}$ (h)	2.93 ± 1.04	1.06 ± 0.07
T_{max} (h)	2.33 ± 1.53	0.08 ± 0.00
C_{max} (ng/mL)	94.47 ± 28.99	13615 ± 838.18
AUC_{0-t} (ng/mL·h)	511.14 ± 140.63	15565.29 ± 2027.28
$AUC_{0-\infty}$ (ng/mL·h)	592.55 ± 57.78	15671.81 ± 2001.37
CL (mL/h/kg)	–	159.01 ± 72.44
MRT (h)	5.04 ± 2.15	1.31 ± 0.07

749.4496 and m/z 101.0241 were the same as those of AB4 (Fig. 3b).

23-HA. 23-HA was the saponin of AB4 with a retention time of 13.33 min ($[M-H]^-$ m/z 471.3480). No fragment ions were detected due to the stable structure of pentacyclic triterpenoids (Fig. 3c).

3.2.2. Metabolite identification in rat plasma, urine, faeces, and bile

A total of 29 metabolites were detected in plasma, urine, faeces, and bile after a single oral dose of AB4 (100 mg/kg) to SD rats. Retention time (Rt), potential molecular formula, and significant fragment ions are presented in Table 2 as the essential details for all metabolites. Excellent mass accuracy was demonstrated by the mass errors between the actual and theoretical masses, which ranged from 3.7 ppm to 4.7 ppm (1×10^{-6}). The possible structures of these metabolites were identified in accordance with their fragments, and the hypothesized metabolic routes are depicted in Fig. 4 (plasma, urine, and faeces) and Fig. 5 (bile). The extracted ion current chromatograms of these metabolites were respectively shown in Supplementary Material Fig. S1.

3.2.2.1. Plasma and urine. Except for the parent AB4, no significant metabolite was detected in plasma and urine.

3.2.2.2. Faeces. In faeces, a total of nine metabolites were detected: AA3, 23-HA, M1074-1, M1074-2, M942, M928, M912, M766, and M604.

M1074-1, M1074-2. The retention times of M1074-1 and M1074-2 were 8.94 min and 9.12 min, respectively, and the molecular ion ($[M + HCOO]^-$) m/z 1119.5593 was obtained in full-scan MS. According to the exact mass, M1074-1 and M1074-2 shared the same molecular formula of $C_{53}H_{86}O_{22}$. The major fragment ions of M1074-1 at m/z 469.1563, 247.0817, 161.0437, and 101.0229 were the same as those of AB4 (Fig. 6a). The major fragment ions of M1074-2 at m/z 749.4499, 323.1010, 161.0456, and 101.0234 were the same as AB4 (Fig. 6b). According to the ion fragments of the two metabolites, it was speculated that M1074-1 was the metabolite produced by AB4 losing the sugar ring B1, while M1074-2 was the metabolite produced by AB4 losing the sugar ring A1. Based on mass differences, both M1074-1 and M1074-2 were

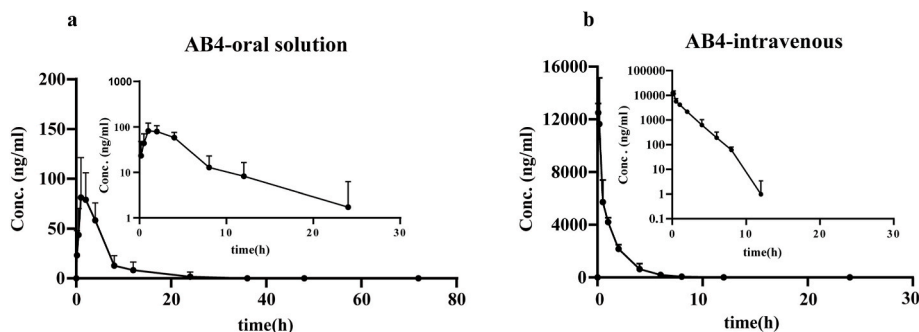


Fig. 2. Mean plasma concentration-time profiles of AB4 after oral and intravenous administration. (a) oral solution (122 mg/kg) and (b) intravenous (1.22 mg/kg). The plasma concentration of AB4 in the inner figures was converted to logarithmic data.

Table 2
Information on AB4 metabolites detected in rat plasma, urine, faeces, and bile by using UHPLC-Q Exactive Plus MS.

ID	Pathway	Formula	RT	Measured [M-H] ⁻ <i>m/z</i>	Measured [M + FA] ⁻ <i>m/z</i>	Mass error (ppm) [#]	Fragment ions		Source
							[M-H] ⁻	[M + FA] ⁻	
AB4	–	C ₅₉ H ₉₆ O ₂₆	8.85		1265.6182	0.8		749.4473,469.1569,323.0974,247.0824,161.0437,101.0231	F, U, P
AA3	M912-[Glu]	C ₄₁ H ₆₆ O ₁₂	12.21		795.4569	4.2		749.4472,603.3899,585.3797,471.3494,101.0229	F
23-HA	M604-[Ara]	C ₃₀ H ₄₈ O ₄	13.43	471.3499		4.0	471.3513		F, B
M1074-1	AB4-[Rha]	C ₅₃ H ₈₆ O ₂₂	8.94		1119.5633	3.6		603.3901,469.1563,247.0817,161.0437,101.0229	F
M1074-2	AB4-[Rha]	C ₅₃ H ₈₆ O ₂₂	9.12		1119.5643	4.7		749.4499,603.3881,323.1010,161.0456,101.0234	F
M942	M1074-1-[Ara]	C ₄₈ H ₇₈ O ₁₈	10.13		987.5203	3.3		471.3509,469.1570,101.0230	F
M928	M1074-2-[Rha]	C ₄₇ H ₇₆ O ₁₈	9.26		973.5049	3.4		603.3903,323.0975	F
M912	M1074-2-[Glu]	C ₄₇ H ₇₆ O ₁₇	10.35		957.5090	2.6		791.4550,749.4479,101.0232	F
M766	M942-[Glu]/M912-[Rha]	C ₄₁ H ₆₆ O ₁₃	10.56		811.4515	3.7		603.3904	F
M664-1	M648+[O]	C ₃₆ H ₅₆ O ₁₁	8.57	663.3760		1.5	663.3768,487.3437,175.0241		B
M664-2	M648+[O]	C ₃₆ H ₅₆ O ₁₁	9.36	663.3761		1.7	663.3755,487.3437,175.0240		B
M662-1	M664-1/M664-2-[2H]	C ₃₆ H ₅₄ O ₁₁	9.83	661.3593		0.0	661.3616,485.3286,441.3380,175.0242		B
M662-2	M664-1/M664-2-[2H]	C ₃₆ H ₅₄ O ₁₁	10.55	661.3602		1.4	661.3608,485.3293,175.0257		B
M648	23-HA+[GluA]	C ₃₆ H ₅₆ O ₁₀	11.63	647.3818		2.6	647.3812,471.3489,175.0237		B
M646	M648-[2H]	C ₃₆ H ₅₄ O ₁₀	11.96	645.3657		2.0	645.3669,469.3333,439.3222,175.0240		B
M636-1	M618+[O]+[2H]	C ₃₅ H ₅₆ O ₁₀	10.89	635.3813		1.9	635.3815,459.3490,175.0242		B
M636-2	M618+[O]+[2H]	C ₃₅ H ₅₆ O ₁₀	11.34	635.3813		1.9	635.3813,459.3490,175.0241		B
M636-3	M618+[O]+[2H]	C ₃₅ H ₅₆ O ₁₀	12.35	635.3817		2.5	635.3812,459.3543,175.0259		B
M618	23-HA-[COOH]+[O]-[2H]+[GluA]	C ₃₅ H ₅₄ O ₉	12.51	617.3713		2.9	617.3707,411.3382,175.0242		B
M616-1	M618-[2H]	C ₃₅ H ₅₂ O ₉	13.11	615.3557		2.9	615.3694,439.3226,175.0242		B
M616-2	M618-[2H]	C ₃₅ H ₅₂ O ₉	13.23	615.3548		1.5	615.3588,439.3227,175.0241		B
M604	M750-[Rha]/M766-[Glu]	C ₃₅ H ₅₆ O ₈	12.40	603.3925		3.8	603.3908		F
M584-1	M568-1/M568-2+[O]	C ₃₀ H ₄₈ O ₉ S	6.55	583.2948		0.3	583.2963,553.2836,537.2891,79.9563		B
M584-2	M568-1/M568-2+[O]	C ₃₀ H ₄₈ O ₉ S	7.58	583.2957		1.9	583.2974,553.2880,537.2515,79.9564		B
M568-1	23-HA+[O]+[SO ₃]	C ₃₀ H ₄₈ O ₈ S	7.60	567.3002		0.9	567.3016,96.9587,79.9565		B
M568-2	23-HA+[O]+[SO ₃]	C ₃₀ H ₄₈ O ₈ S	8.09	567.2996		-0.2	567.3008,96.9597,79.9563		B
M548-1	M546+[2H]	C ₃₂ H ₅₂ O ₇	11.78		593.3715	3.4		593.3711	B
M548-2	M546+[2H]	C ₃₂ H ₅₂ O ₇	11.98		593.3705	1.7		593.3707,575.3577	B
M546	23-HA+[CH ₂ CO]+2[O]	C ₃₂ H ₅₀ O ₇	12.22		591.3558	3.2		591.3551,573.3497	B
M486	23-HA+[O]-[2H]	C ₃₀ H ₄₆ O ₅	12.23	485.3288		3.3	485.3292,441.3392		B

Notes: “P”: plasma; “U”: urine; “F”: faeces; “B”: bile.

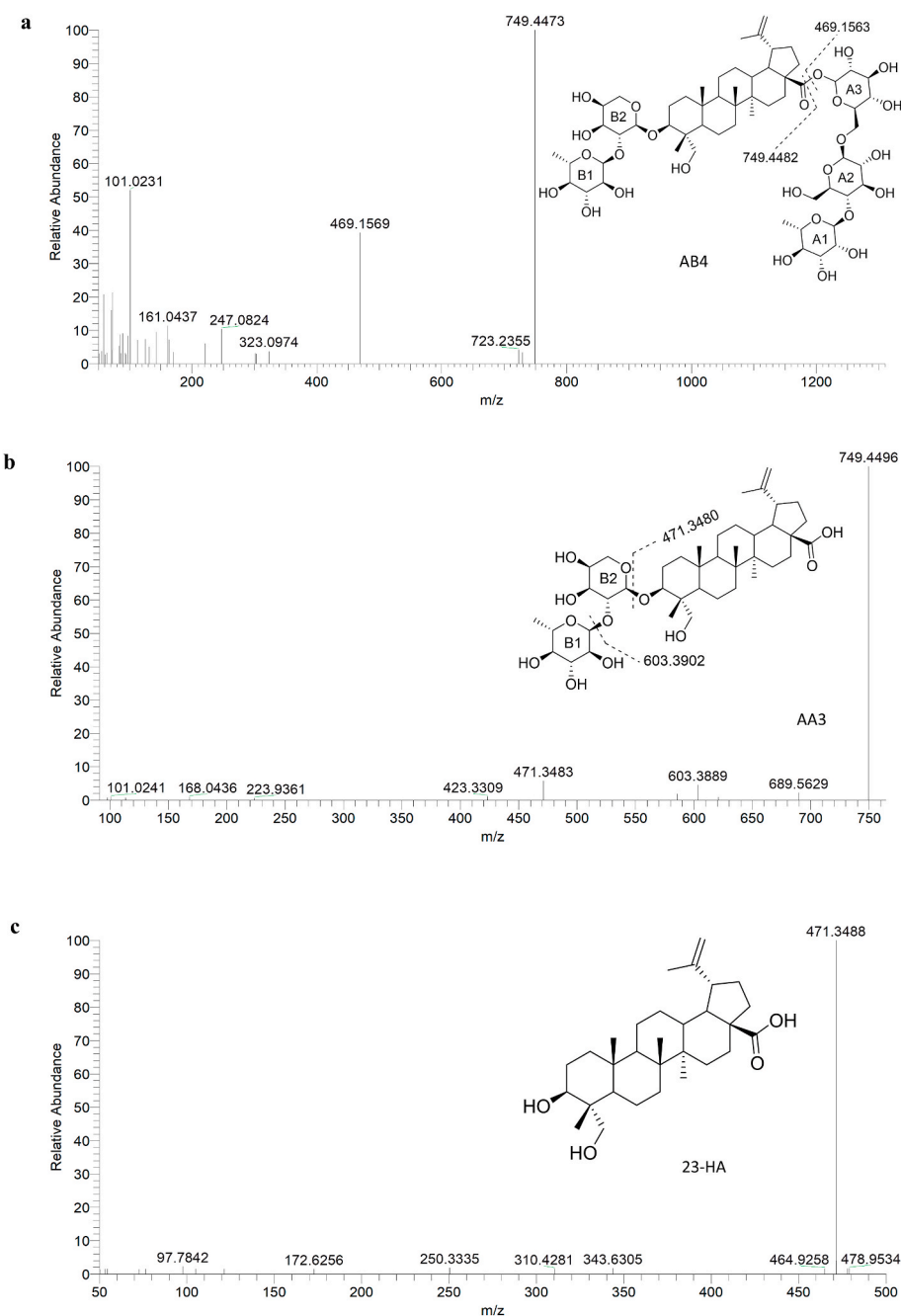


Fig. 3. The mass spectra and proposed fragmentation of (a) AB4, (b) AA3, and (c) 23-HA.

deglycosylation products of AB4.

M928. The retention time of M928 was 9.26 min, and the molecular ion ($[M + HCOO]^-$) m/z 973.5014 was obtained in full-scan MS. According to the exact mass, the molecular formula of M928 was $C_{47}H_{76}O_{18}$. The major fragment ions at m/z 323.0975 were the same as AB4. M928 was tentatively identified as the deglycosylation product of AB4 via the loss of sugar ring A1 and sugar ring B1 moieties (Fig. 6c).

M942. The retention time of M942 was 10.13 min, and the molecular ion ($[M + HCOO]^-$) m/z 987.5170 was obtained in full-scan MS. According to the exact mass, the molecular formula of M942 was $C_{48}H_{78}O_{18}$. The major fragment ions at m/z 469.1570 and 101.0230 were the same as those of AB4. M942 was tentatively identified as the deglycosylation product of AB4 via the loss of the sugar ring B moiety based on the mass difference (Fig. 6d).

M912. The retention time of M912 was 10.35 min, and the molecular

ion ($[M + HCOO]^-$) m/z 957.5065 was obtained in full-scan MS. According to the exact mass, the molecular formula of M912 was $C_{47}H_{76}O_{17}$. The major fragment ions at m/z 749.4479 and 101.0232 were the same as AB4. M912 was tentatively identified as the deglycosylation product of AB4 via the loss of sugar rings A1 and A2 moieties (Fig. 6e).

M766. The retention time of M766 was 10.56 min, and the molecular ion ($[M + HCOO]^-$) m/z 811.4485 was obtained in full-scan MS. According to the exact mass, the molecular formula of M766 was $C_{41}H_{66}O_{13}$. The major fragment ions at m/z 603.3904 were the same as AA3. M766 was tentatively identified as the deglycosylation product of AB4 via the loss of sugar rings A1 and A2 and sugar ring B1 moieties (Fig. 6f).

M604. The retention time of M604 was 12.40 min, and the molecular ion ($[M-H]^-$) m/z 603.3902 was obtained in full-scan MS. According to

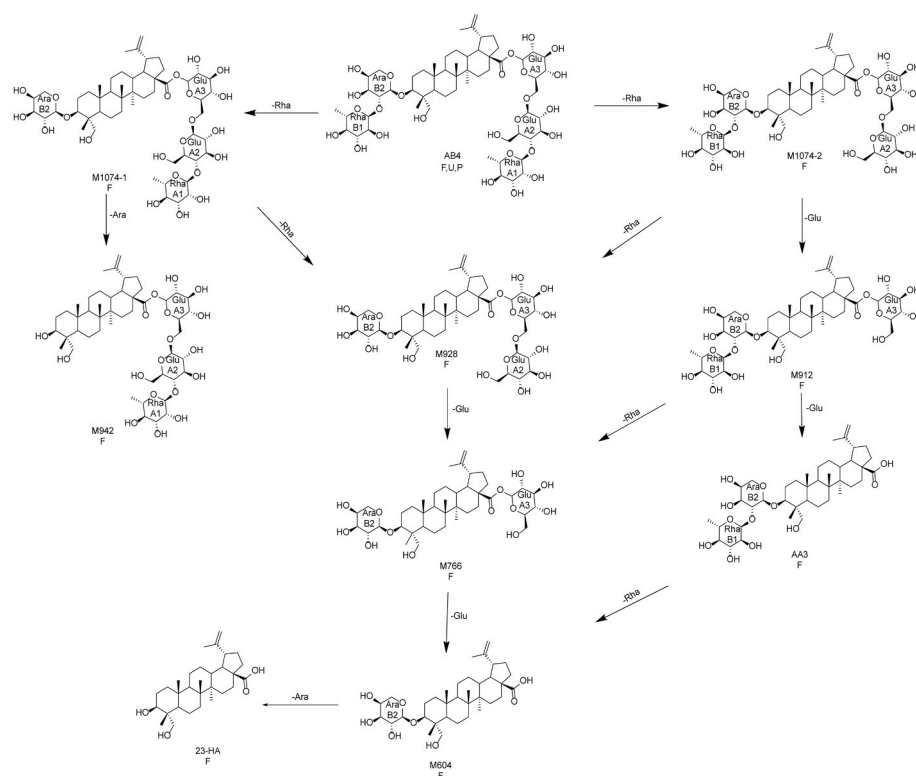


Fig. 4. Pathway of AB4 metabolism in the plasma, urine, and faeces of SD rats.

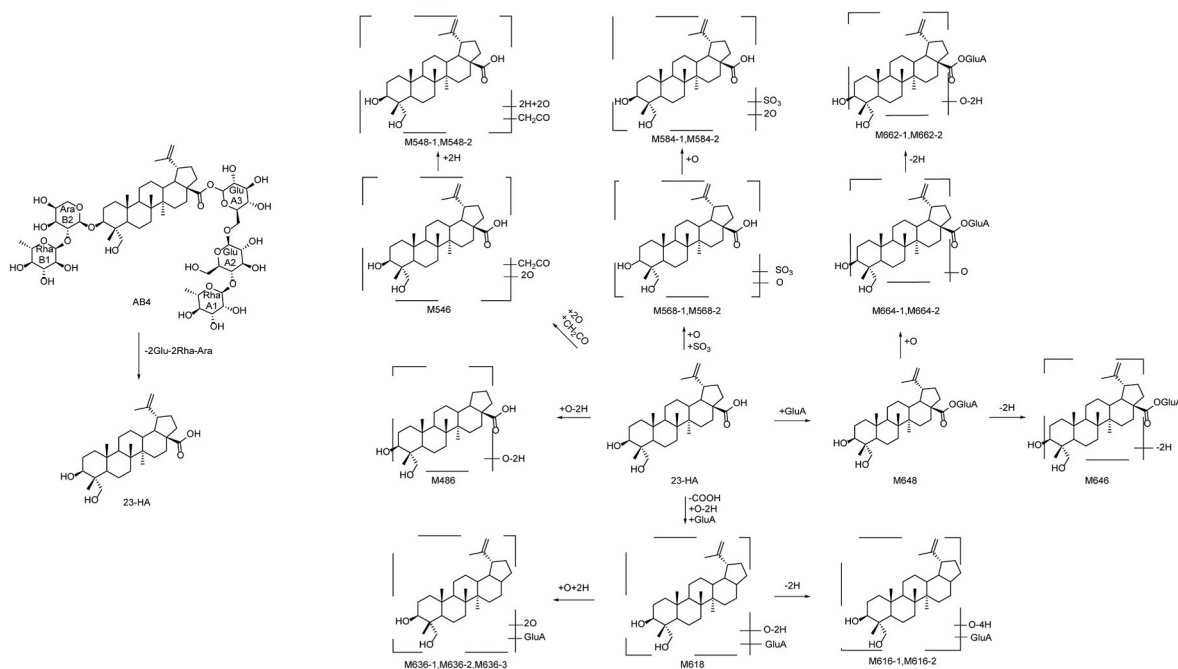


Fig. 5. Pathway of AB4 metabolism in the bile of SD rats.

the exact mass, the molecular formula of M604 was $C_{35}H_{56}O_8$. The major fragment ions at m/z 603.3908 were the same as AA3. M604 was tentatively identified as the deglycosylation product of AB4 via the loss of sugar ring A and sugar ring B1 moieties (Fig. 6g).

3.2.2.3. Bile. In bile, a total of 20 metabolites were detected, and the major metabolites are described as follows.

M664-1, M664-2. Two isomers of M664-1 and M664-2 ($C_{36}H_{56}O_{11}$)

were simultaneously extracted at retention times of 8.57 min and 9.36 min with m/z 663.3750 [M-H]⁻, respectively. The main fragment at m/z 487.3437 was 176.0313 Da less than the molecular ion and 15.9957 Da more than the molecular ion of 23-HA, suggesting that the two metabolites were oxidation metabolites of M648 at two different positions (Fig. 7a).

M648. The retention time of M648 was 11.63 min, and the molecular ion ([M-H]⁻) m/z 647.3801 was obtained in full-scan MS. According to

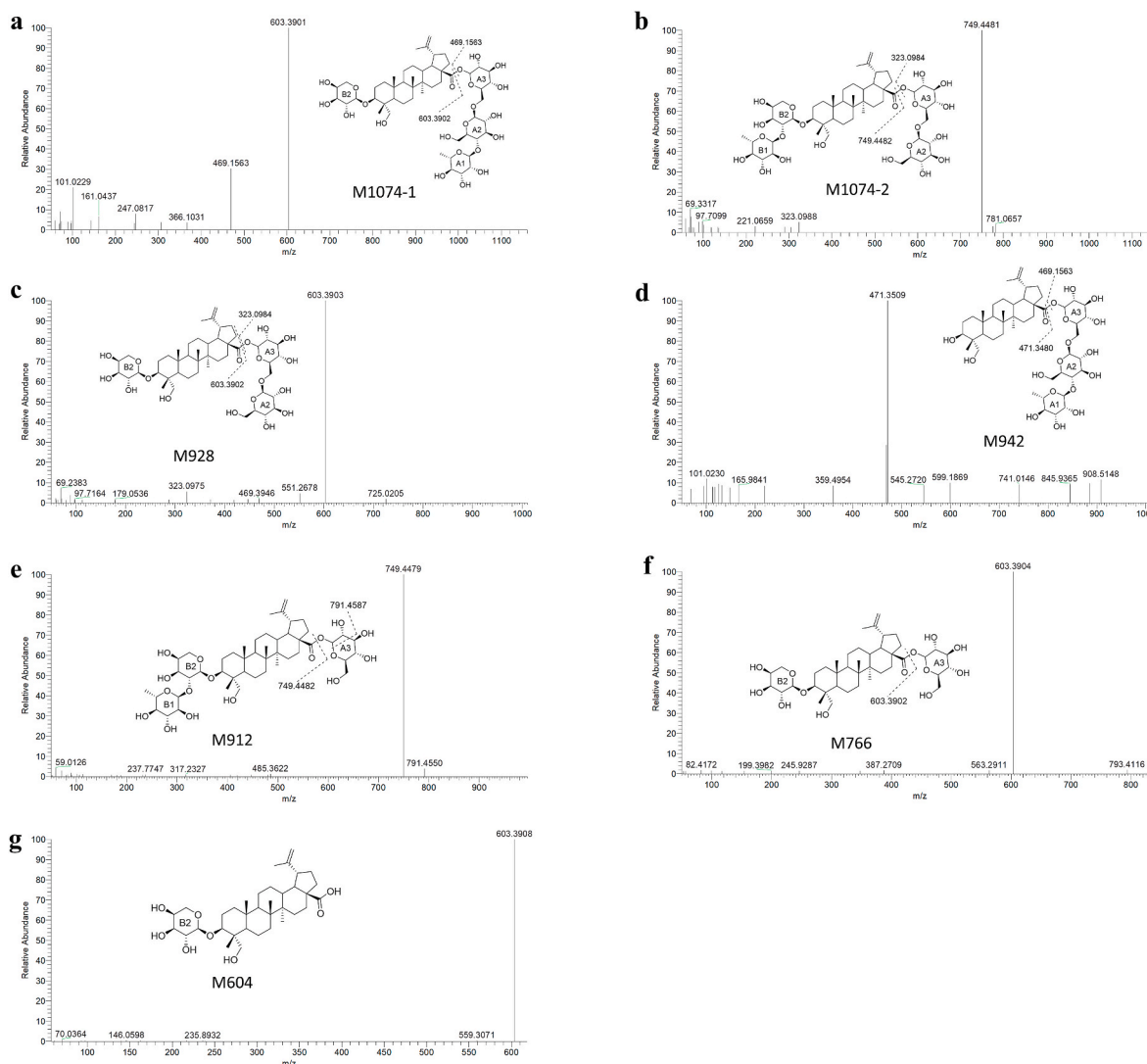


Fig. 6. The mass spectra and proposed fragmentation of (a) M1074-1, (b) M1074-2, (c) M928, (d) M942, (e) M912, (f) M766, and (g) M604.

the exact mass, the molecular formula of M648 was $C_{36}H_{56}O_{10}$. The main fragment at m/z 471.3489 was the same as 23-HA and 176.0313 Da less than the molecular ion, suggesting that M648 was the glucuronide metabolite of 23-HA (Fig. 7b).

M618. The retention time of M618 was 12.51 min, and the molecular ion ($[M-H]^-$) m/z 617.3695 was obtained in full-scan MS. According to the exact mass, the molecular formula of M618 was $C_{35}H_{64}O_9$. The main fragment at m/z 441.3382 was 176.0313 Da less than the molecular ion and 43.9904 Da less than the fragment at m/z 485.3286 of M662-1 and M662-2, suggesting that M618 was a decarboxylation, oxidative dehydrogenation and glucuronide metabolite of 23-HA (Fig. 7c).

M584-1, M584-2. Two isomers of M584-1 and M584-2 ($C_{30}H_{48}O_9S$) were simultaneously extracted at retention times of 6.55 min and 7.58 min with m/z 583.2946 $[M-H]^-$, respectively. The main fragment was m/z 79.9564, and the molecular ion was 15.9949 Da more than the molecular ions of M568-1 and M568-2, suggesting that the two metabolites were oxidation metabolites of M568-1 and M568-2 (Fig. 7d).

M568-1, M568-2. Two isomers of M568-1 and M568-2 ($C_{30}H_{48}O_8S$) were simultaneously extracted at retention times of 7.60 min and 8.09 min with m/z 567.2997 $[M-H]^-$, respectively. The main fragments at m/z were 96.9587 and 79.9565, suggesting that the two metabolites were oxidation and sulfation metabolites of 23-HA (Fig. 7e).

Other metabolites are listed in Fig. S2 in the Supplementary material.

3.3. Comparison of the PK properties of AB4 in rectal suppository and enteric capsule formulations

To improve the exposure of the parent drug of AB4 and its potential active metabolites in the circulatory system and to meet future clinical needs, we developed an AB4 suppository and AB4 enteric capsule and investigated their PK properties. The plasma concentration-time profiles of AB4 for the suppository and enteric capsule are shown in Fig. 8. The calculated PK parameters are tabulated in Table 3.

The results show that AB4 could be rapidly absorbed in vivo and reached C_{max} within 1.3 h after rectal administration of the suppository and oral administration of the AB4 enteric capsule. The C_{max} values for the rectal suppository and enteric capsule were 10394.16 ± 5041.38 ng/mL and 219.96 ± 103.32 ng/mL, respectively. Moreover, the $T_{1/2}$ of rectal suppository and enteric capsule administration were 6.93 ± 0.95 h and 2.82 ± 1.29 h, respectively. As shown in Table 3, the $AUC_{0-\infty}$ after rectal administration of suppository and oral administration of AB4 enteric capsules were 29115.16 ± 14345.48 ng/mL·h and 1049.00 ± 101.54 ng/mL·h, respectively. The AUC of AB4 after rectal administration of the suppository was 49- and 28-fold higher than that after oral administration of AB4 solution and AB4 enteric capsules.

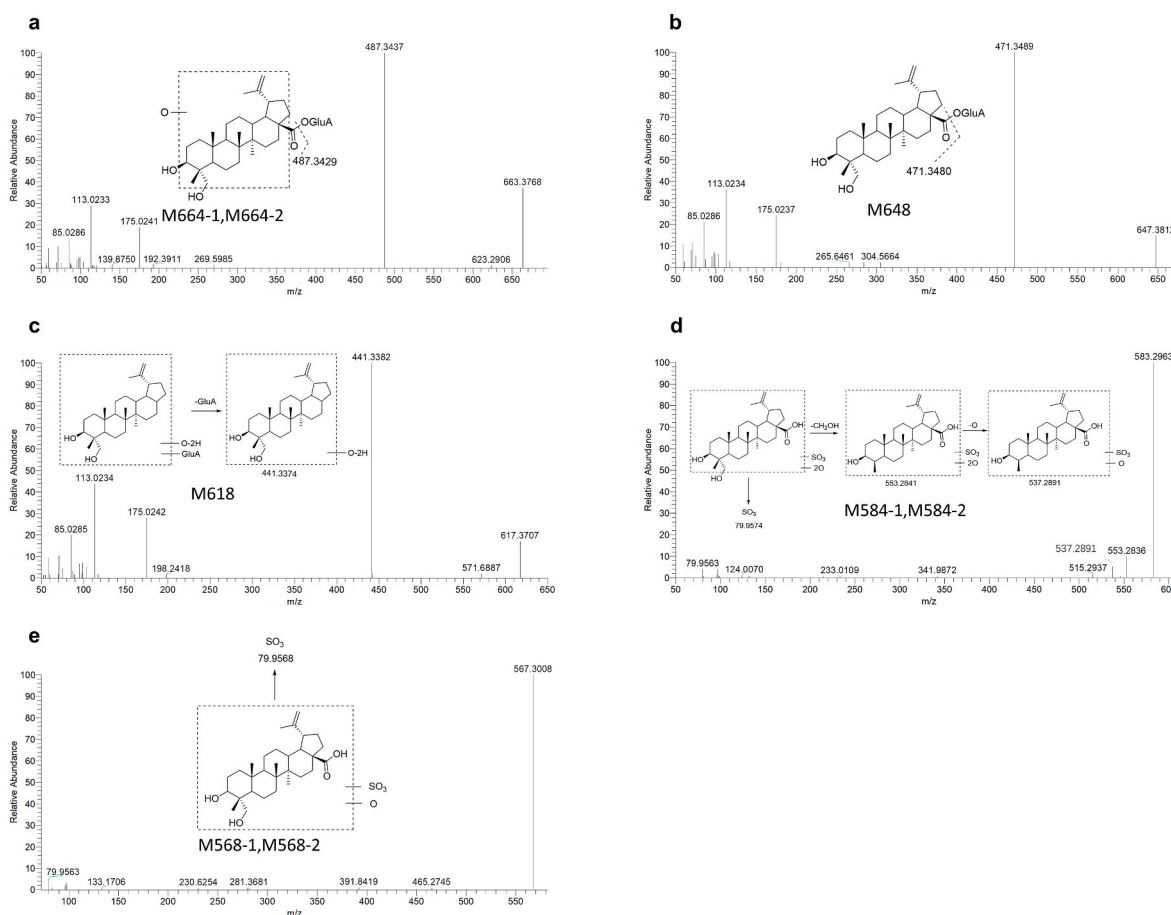


Fig. 7. The mass spectra and proposed fragmentation of (a) M664–1, M664–2, (b) M648, (c) M618, (d) M584–1, M584–2, and (e) M568–1, M568–2.

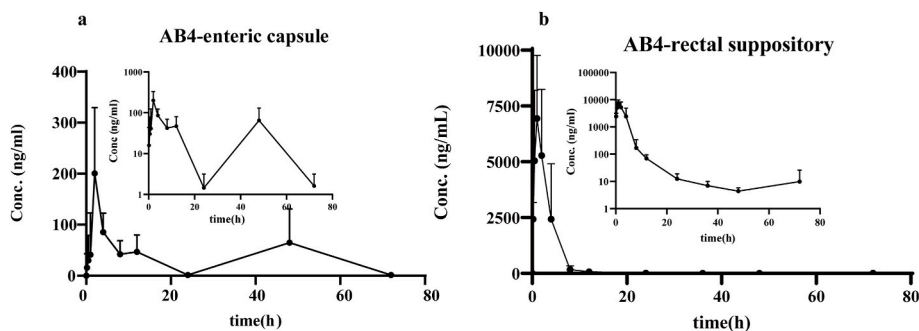


Fig. 8. Mean plasma concentration-time profiles of AB4 after oral administration of AB4 enteric capsule and rectal administration of suppository in rats ($n = 7$). (a) enteric capsule (122 mg/kg) (b) rectal suppository (122 mg/kg). The plasma concentration of AB4 in the inner figures was converted to logarithmic data.

Table 3

Pharmacokinetic parameters of AB4 after rectal administration of suppository and oral administration of enteric capsules. ($n = 7$, mean $\pm$ SD).

Parameters	AB4-rectal suppository (122 mg/kg)	AB4-enteric capsule (122 mg/kg)
$T_{1/2}$ (h)	6.93 ± 0.95	2.82 ± 1.29
T_{max} (h)	1.30 ± 0.67	1.15 ± 0.57
C_{max} (ng/mL)	10394.16 ± 5041.38	219.96 ± 103.32
AUC_{0-t} (ng/mL·h)	28944.91 ± 14497.25	1025.64 ± 97.23
$AUC_{0-\infty}$ (ng/mL·h)	29115.16 ± 14345.48	1049.00 ± 101.54
MRT (h)	3.60 ± 1.67	6.05 ± 2.34

4. Discussion

With the application of HPLC-QQQ-MS/MS, we successfully evaluated the concentration of AB4 in rat plasma after oral and intravenous administration. AB4 could be rapidly absorbed and eliminated in vivo after oral administration, with T_{max} and $T_{1/2}$ at 2.93 h and 2.33 h, respectively. As shown in Table 1, the C_{max} and AUC of AB4 after oral administration were quite low, which meant that the absorption characteristics and bioavailability of AB4 after oral administration were quite poor. However, despite its low bioavailability after oral administration, AB4 has been found to protect against chronic kidney injury (CKI) induced by adenine in rats after oral administration (Lei et al., 2019). Additionally, our recent study indicated that oral administration of AB4 showed significant anti-hyperuricaemic effects in mice. We

proposed that this inconsistency in the exposure-response relationship of AB4 was attributed to potential active metabolites. Thus, we identified metabolites of AB4 in rat urine, blood, faeces, and bile after oral administration.

In the development of new drugs, it is necessary to determine the potential metabolic pathways *in vivo*, which can be used to explain the efficacy or safety issues. UHPLC-Q Exactive Plus MS was used to study the metabolism of AB4 in the pooled plasma during 0–24 h and in the urine, faeces, and bile during 0–72 h. Our findings fully gave crucial insight into the knowledge of the *in vivo* metabolic pathways of AB4 following oral administration, despite the fact that the structures of several metabolites were highly challenging to identify using mass spectrometry alone. A total of 29 AB4 metabolites were identified from rat biological samples, including two urine metabolites, nine faecal metabolites, and 20 bile metabolites. Deglycosylation, oxidation, dehydrogenation, reduction, sulfation, hydration, acetylation, and glucuronidation were established as the main metabolic pathways for AB4 by UHPLC-Q Exactive Plus MS. Due to the lack of standards of metabolites, the concentration of metabolites in different biological samples could not be determined. No significant metabolites were detected in the plasma and urine, only the parent drug AB4, but the metabolites were detected in the bile. Our lab's previous study (data not published) has found that AB4 is unstable in intestinal contents and can be metabolized by intestinal flora or enzymes, with AA3 and 23-HA as two metabolites in the intestine. The metabolite AA3 can be further metabolized to 23-HA in the intestine content. This finding is consistent with the *in vitro* study previously published (Wan et al., 2017). Furthermore, we found that AB4 is stable in rat primary hepatocytes and 23-HA can be extensively metabolized in hepatocytes, which is consistent to the results that abundant 23-HA related substance are found in bile and only AB4 is found in plasma. The metabolites in bile were not detected in the faeces possibly due to the detection limit, and the metabolites were metabolized again in the intestinal tract. Notably, relevant studies found that AB4 metabolites could be produced by intestinal bacteria, such as AA3, 23-HA, M1074–1, M1074–2, M942, M928, M912, M766, and M604 (Wan et al., 2017). Most of the metabolites detected in bile were associated with 23-HA. Combined with our previous studies (Guo et al., 2018), we found that after oral administration of AB4, AB4 was metabolized into 23-HA in large quantities before entering the liver. Although 23-HA was highly exposed in the portal vein blood before entering the liver, 71% of 23-HA was cleared after entering the liver due to extensive metabolism in the liver; therefore, most of the metabolites detected in bile were associated with 23-HA. Overall, the oral dosage of AB4 underwent extensive biotransformation in both intestine and liver.

Given the extremely low bioavailability, we also developed two formulations, the rectal suppository and enteric capsule, to increase AB4 exposure *in vivo*. However, the PK results showed that only the rectal suppository largely increased the exposure of AB4 compared to the oral formulation of the saline solution or the enteric capsule, with 49 and 28 times higher AUC values. In addition, the $T_{1/2}$ of AB4 was extended to approximately 7 h after rectal administration but was not improved in the enteric capsule. The slight increase in the AUC in the enteric capsule group indicated that the reason for the low oral bioavailability of AB4 was not caused from its instability in gastric acid but from metabolism by the intestinal bacteria and the first-pass effect of the liver. Hence, the rectal suppository group, which has a minimal first-pass effect on the liver, significantly increased the stability of AB4 in rats. These results were possible due to the advantages of the route of rectal formulation, such as the retention of higher drug quantities, protection of enzymatically unstable drugs, increased effectiveness, and minimal first-pass drug metabolism (Ozgüney et al., 2014; Purohit et al., 2018). After rectal suppository administration, AB4 is absorbed from the lower rectum and enters directly into the systemic circulation via the middle and inferior haemorrhoidal veins (Bialik et al., 2021); this prevents drug metabolism (or hydrolysis) in the gastrointestinal tract due to the presence of numerous metabolic enzymes in the gastrointestinal tract.

Basically, a rectal suppository can be an alternative formulation of AB4 to obtain both higher and longer exposure, which can not only increase the exposure of AB4 *in vivo* but also be suitable for chronic diseases that need long-term administration.

As a saponin component of *Pulsatilla chinensis*, 23-HA has attracted interest due to its pharmacological activities, such as antitumour (Zhang et al., 2015) and against doxorubicin-induced cardiotoxicity (Zhou et al., 2015). In the present study, we found that most of the metabolites detected in bile were associated with 23-HA; we hypothesized that 23-HA might be the representative active metabolite of AB4. Hence, we detected the concentration of 23-HA in SD rat plasma after rectal administration of the AB4 suppository, oral administration of the AB4 solution, and oral administration of the enteric capsule at a dose of 122 mg/kg. The calculated plasma concentration-time profiles (Fig. S3) and PK parameters (Table S2) are shown in the Supplementary material. The results showed that 23-HA in the enteric capsule group had the highest 23-HA AUC. 23-HA in the rectal suppository group was much lower possibly due to the lack of enzymes in the rectum, causing difficulty to metabolize AB4 to 23-HA. Among the three formulations (solution, enteric capsule, and rectal suppository), rectal suppositories could greatly increase the exposure of the AB4 parent *in vivo*, while enteric-coated capsules could increase the content of the main metabolite 23-HA in systemic plasma. However, according to the present data, we cannot conclude whether the parent (AB4) or its metabolites exert higher activities; further pharmacological experiments are needed for verification.

We comprehensively investigated the PK and metabolism profiles of AB4 in different administrative routes and different formulations. Based on AUC exposure *in vivo*, the stability of AB4 was ordered as follows: rectal suppository > enteric capsule > saline solution. However, extensive metabolism could moderately contribute to the biological response of AB4 after oral administration. In our future studies, further comparison of the pharmacological activity between these formulations will be conducted, which will provide for a better understanding of the new AB4 formulations.

5. Conclusions

In the present study, we distinctly investigated the PK properties and the absolute bioavailability of AB4 after oral administration, which were extremely low. This led to further investigations of the metabolic fate of AB4. A comprehensive metabolite identification for AB4 was conducted, and the results updated the metabolic profiles and the biotransformation pathways of AB4 *in vivo*. Finally, new formulations were developed to meet the clinical needs of AB4 and increase its exposure and lasting time *in vivo*, thus increasing its pharmacological activities. The PK comparison results showed that rectal formulation could increase the exposure and slow down the elimination of AB4. In the future, pharmacological activities will be further compared between the above formulations and administrative routes.

CRediT authorship contribution statement

Yaqing Ye: Conceptualization, Methodology, Validation, Formal analysis, carried out the animal experiments, Writing – original draft, did the pharmacokinetic analysis. **Mingzhen Xue:** Conceptualization, Methodology, Validation, Formal analysis, carried out the animal experiments, Writing – original draft, did the metabolite analysis. **Xiaoting Tian:** Conceptualization, Validation, carried out the animal experiments. **Hongwei Gao:** Conceptualization, Resources. **Pei Hu:** carried out the animal experiments. **Linwei Wang:** carried out the animal experiments. **Jing Leng:** Supervision, Writing – review & editing. **Yaru Xue:** Conceptualization, Supervision, Formal analysis, Funding acquisition, carried out the animal experiments, Writing – review & editing. **Chenggang Huang:** Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116694>.

References

- Bialik, M., Kuras, M., Sobczak, M., Oledzka, E., 2021. Achievements in thermosensitive gelling systems for rectal administration. *Int. J. Mol. Sci.* 22 (11).
- Gong, Q., He, L.-L., Wang, M.-L., Ouyang, H., Gao, H.-W., Feng, Y.-L., Yang, S.-L., Du, L.-J., Li, J., Luo, Y.-Y., 2019. Anemoside B4 protects rat kidney from adenine-induced injury by attenuating inflammation and fibrosis and enhancing podocin and nephrin expression. *Evid Based Compl. Alternat. Med.* 2019, 8031039.
- Guo, X., Xie, Y., Lian, S., Li, Z., Gao, Y., Xu, Z., Hu, P., Chen, M., Sun, Z., Tian, X., Huang, C., 2018. A sensitive HPLC-MS/MS method for the simultaneous determination of anemoside B4, anemoside A3 and 23-hydroxybetulinic acid: application to the pharmacokinetics and liver distribution of *Pulsatilla chinensis* saponins. *Biomed. Chromatogr.* 32 (3).
- He, M., Ouyang, H., He, M., Tan, T., Li, J., Zhang, X., Jia, J., Feng, Y., Yang, S., 2017. Application of a liquid chromatography-tandem mass spectrometry method to the pharmacokinetics, tissue distribution and excretion in the study of anemoside B4, a novel antiviral agent candidate. *Rats Biomed. Chromatogr.* 31 (7).
- Hop, C.E., Wang, Z., Chen, Q., Kwei, G., 1998. Plasma-pooling methods to increase throughput for in vivo pharmacokinetic screening. *J. Pharmaceut. Sci.* 87 (7), 901–903.
- Hu, T., Ge, X., Wang, J., Zhang, N., Diao, X., Hu, L., Wang, X., 2021. Metabolite identification of iridin in rats by using UHPLC-MS/MS and pharmacokinetic study of its metabolite irigenin. *J. Chromatogr., B: Anal. Technol. Biomed. Life Sci.* 1181, 122914.
- Hu, Y., Chen, X., Duan, H., Hu, Y., 2009. Chinese herbal medicinal ingredients inhibit secretion of IL-6, IL-8, E-selectin and TXB2 in LPS-induced rat intestinal microvascular endothelial cells. *Immunopharmacol. Immunotoxicol.* 31 (4), 550–555.
- Hu, Y., He, K., Wang, X., 2016. Role of Chinese herbal medicinal ingredients in secretion of cytokines by PCV2-induced endothelial cells. *J. Immunot.* 13 (2), 141–147.
- Huang, J., Gong, M.J., Bai, J.Q., Su, H., Gong, L., Huang, Z.H., Qiu, X.H., Xu, W., Zhang, J., 2022. Differential metabolic profiles of ginsenosides in artificial gastric juice using ultra-high-pressure liquid chromatography coupled with linear ion trap-Orbitrap mass spectrometry. *Biomed. Chromatogr.* 36 (12), e5493.
- Jin, M.-M., Zhang, W.-D., Xu, Y.-M., Du, Y.-F., Sun, Q., Guo, W., Cao, L., Xu, H.-J., 2017. Simultaneous determination of 12 active components in the roots of *Pulsatilla chinensis* using tissue-smashing extraction with liquid chromatography and mass spectrometry. *J. Separ. Sci.* 40 (6), 1283–1292.
- Lei, F., Gao, H., Li, J., Du, L., He, L., Wang, M., Gong, Q., Wong, S., Yang, S., Yu, X., Luo, Y., Feng, Y., 2019. Proteomic changes in rat kidney injured by adenine and the regulation of anemoside B4. *J. Chin. Pharmaceut. Sci.* 28 (1), 10–20.
- Li, Y.-H., Zou, M., Han, Q., Deng, L.-R., Weinshilboum, R.M., 2020. Therapeutic potential of triterpenoid saponin anemoside B4 from *Pulsatilla chinensis*. *Pharmacol. Res.* 160, 105079.
- Ling, Y., Lin, Z., Zha, W., Lian, T., You, S., 2016. Rapid detection and characterisation of triterpene saponins from the root of *Pulsatilla chinensis* (Bunge) Regel by HPLC-ESI-QTOF-MS/MS. *Phytochem. Anal.* 27 (3–4), 174–183.
- Ouyang, H., Zhou, M., Guo, Y., He, M., Huang, H., Ye, X., Feng, Y., Zhou, X., Yang, S., 2014. Metabolites profiling of *Pulsatilla* saponin D in rat by ultra performance liquid chromatography-quadrupole time-of-flight mass spectrometry (UPLC/Q-TOF-MS/MS). *Fitoterapia* 96, 152–158.
- Ozgüney, I., Kardhiqi, A., Yıldız, G., Ertan, G., 2014. In vitro-in vivo evaluation of in situ gelling and thermosensitive ketoprofen liquid suppositories. *Eur. J. Drug Metab. Pharmacokinet.* 39 (4), 283–291.
- Purohit, T.J., Hanning, S.M., Wu, Z., 2018. Advances in rectal drug delivery systems. *Pharmaceut. Dev. Technol.* 23 (10), 942–952.
- Wan, J.-Y., Zhang, Y.-Z., Yuan, J.-B., Yang, F.-Q., Chen, Y., Zhou, L.-D., Zhang, Q.-H., 2017. Biotransformation and metabolic profile of anemoside B4 with rat small and large intestine microflora by ultra-performance liquid chromatography-quadrupole time-of-flight tandem mass spectrometry. *Biomed. Chromatogr.* 31 (5).
- Zhang, D.-M., Xu, H.-G., Wang, L., Li, Y.-J., Sun, P.-H., Wu, X.-M., Wang, G.-J., Chen, W.-M., Ye, W.-C., 2015. Betulinic acid and its derivatives as potential antitumor agents. *Med. Res. Rev.* 35 (6), 1127–1155.
- Zhou, F., Hao, G., Zhang, J., Zheng, Y., Wu, X., Hao, K., Niu, F., Luo, D., Sun, Y., Wu, L., Ye, W., Wang, G., 2015. Protective effect of 23-hydroxybetulinic acid on doxorubicin-induced cardiotoxicity: a correlation with the inhibition of carbonyl reductase-mediated metabolism. *Br. J. Pharmacol.* 172 (23), 5690–5703.
- Zhou, M., Chen, L., Honghui, H.U., Luo, Y., Fang, C., Zhang, N., 2019. Protective effect of *Pulsatilla* saponin B4 on lipopolysaccharide-induced acute lung injury. *Tradit. Chin. Drug Res. Clin. Pharmacol.*